

Progress Report — Week 3

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Reporting period

15 May 2026 – 21 May 2026

Hours worked

Date	Day	Hours	Activity summary
15 May 2026	Fri	8	CI subconstruct formulas in metrics.py
18 May 2026	Mon	8	Decision quality, trust calibration, effectiveness metrics
19 May 2026	Tue	8	AI allocation and shared cognition modules
20 May 2026	Wed	8	Research paper survey — 30 CI/agile papers identified
21 May 2026	Thu	8	Streamlit single-run UI, with-AI vs without-AI comparison
Total		40	

Work completed

- Implemented collective intelligence subconstructs in `metrics.py`: transactive memory, shared attention, shared reasoning, social sensitivity, participation balance, team engagement, skill diversity.
 - Added decision quality, trust calibration, and Hackman-style team effectiveness (task output, viability, sustainability).
 - Built `ai_support.py` with AI task allocation, shared cognition dashboard, and trust learning.
 - Started systematic analysis of ~30 research papers on CI and agile performance.
 - Created initial Streamlit app with parameter controls and single-run results view.
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Connection to learning objectives

- Applied statistics/research literacy to identify relevant literature and map constructs to code.
 - Demonstrated independent implementation of a non-trivial simulation model.
 - Began connecting academic theory with practical software design.
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Challenges and how they were handled

Translating abstract CI constructs into formulas required explicit assumptions. I documented each formula's purpose and marked weights as simulation assumptions rather than empirical coefficients.

Plan for next week (22–28 May 2026)

- Complete research paper analysis and draft presentation report.
 - Review supervisor feedback on missing process criteria and team engagement.
 - Plan implementation of effort management, skills/knowledge coordination, and task strategy variables.
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Student signature: Arslan Sultan — Date: 22 May 2026

Supervisor acknowledgment: Juan Garbajosa